

CLASS DESCRIPTIONS

The following descriptions explain the purpose, important attributes, and operations of the major classes in the Movie Theatre Website. UUID values are unique identifiers. DateTime represents a date and time, LocalDate represents a calendar date, Decimal represents currency or other precise numeric values, and List<T> represents a collection of values of type T.

DOMAIN CLASSES

Movie

Purpose: Represents a movie that can be displayed in the customer catalogue and scheduled for theatre showings.

Attributes: movieId: UUID identifies the movie; title: String stores the movie name; description: String stores the synopsis; rating: String stores the content rating; runtimeMinutes: Integer stores the runtime; releaseDate: LocalDate stores the release date; posterUrl and trailerUrl: String store media links; status: MovieStatus identifies whether the movie is published, unpublished, or archived.

Operations: validate(): ValidationResult checks required fields and valid values; publish(), unpublish(), and archive() change the movie status; updateDetails(details: MovieDetails): void changes editable movie information; listShowtimes(date: LocalDate): List<Showtime> returns showtimes for a date.

Theatre

Purpose: Represents the theatre location and its seating configuration.

Attributes: theatreId: UUID identifies the theatre; name: String stores its name; seatingArrangement: String describes the layout; cleaningTimeMinutes: Integer stores the cleaning time required between showings; active: Boolean indicates whether the theatre can be scheduled.

Operations: listSeats(): List<Seat> returns the theatre's seats; isAvailable(start: DateTime, end: DateTime): Boolean checks whether the theatre can be used during a period; hasScheduleConflict(start: DateTime, end: DateTime): Boolean checks overlapping showtimes and cleaning time; updateDetails(...): void updates theatre information.

Seat

Purpose: Represents one physical seat in a theatre.

Attributes: `seatId`: UUID identifies the seat; `theatreId`: UUID identifies its theatre; `rowLabel`: String and `seatNumber`: Integer identify its location; `seatType`: `SeatType` identifies the category of seat; `accessible`: Boolean indicates accessibility; `active`: Boolean indicates whether it can be sold.

Operations: `getDisplayLabel()`: String returns a label such as A10;
`canBeSold()`: Boolean checks whether the seat is active and eligible for sale;
`getStatus(showtimeId: UUID)`: `SeatStatus` returns the seat's current status for a showtime.

Showtime

Purpose: Represents a scheduled showing of a movie in a theatre.

Attributes: `showtimeId`, `movieId`, `theatreId`, and `priceScheduleId`: UUID link the showtime to its records; `startTime`, `endTime`, `saleStartTime`, and `saleEndTime`: `DateTime` define the showing and sales period; `status`: `ShowtimeStatus` stores the current state.

Operations: `validate()`: `ValidationResult` checks time and relationship rules;
`publish()`: void makes the showing available for customers; `cancel(reason: String)`: void cancels it and records the reason; `hasConflict(other: Showtime, cleaningTimeMinutes: Integer)`: Boolean checks schedule overlap;
`getAvailableSeats()`: `List<Seat>` returns seats that can be selected;